

Project report

in 260035 LP Laboratory: Computational Statistical Mechanics

Titel | Title

T.b.d but something Flow Ellipsoids

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Wien | Vienna, 2026

Studienkennzahl lt. Studienblatt | degree
programme code as it appears on the student
record sheet:

UA 066 876

Studienrichtung lt. Studienblatt | degree
programme as it appears on the student record
sheet:

Masterstudium Physics

Abstract:

This report presents work done in the 260035 LP Laboratory: Computational Statistical Mechanics, towards a larger project on recreating the experimental findings of Queiros Campos et al (2024) [1] in simulation.

The report includes two different methods of creating densely packed raspberry ellipsoid particles, as well as a qualitative analysis of the magnetic behaviour obtained by applying the Thermal Stoner-Wohlfarth model by Mostarac (2025) [2] to them.

!!! TODO

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1 Introduction

1.1 Motivation

Colloidal motors refer to systems in which colloidal particles, that is, particles evenly dispersed in a carrier medium, are able to travel in a directional manner. Chen et al. (2019) [3]

Treatment of blood clots via thrombolytic drug delivery, has the problem that in the vicinity of a blood clot, there is little flow, by which to efficiently deliver a drug.

One proposed method to improve this treatment is to use magnetic colloidal motors achieved by a rotating magnetic field to induce a recirculation flow in the vicinity, in order to aid the diffusion of the drug or even mechanically erode the clot. See e.g. Cheng et al. (2014). [4]

1.2 Studied System

The system of interest here, is an experimental setup by Queiros Campos (2024) [1]. It is shown in Figure 1. Simplified, the system is initially composed of a water based solution inside a channel of width 1 mm with maghemite nanoparticles of size 20nm and some primary aggregates of them with size 120-140 nm.

A magnetic field is applied, with a gradient in y direction and a rotating component. The rotating field leads to secondary aggregation into the visible rod shaped structures which spin synchronously with the field. The field gradient causes the spinning rods to translate towards the back wall and thus produces the visible gradient in their concentration.

The named parameters include, field amplitude and frequency $H_0 = 6.4 \text{ kA m}^{-1}$, $f = 5 \text{ Hz}$ and the volume fraction of particles $\phi_p = 3.2 \cdot 10^{-3}$.

The paper goes on, to describe how the spinning rods exert a rightward force on the back wall (and adjacent fluid), and therefore experience a net force and translation to the left.

As a result, the fluid in the region with the aggregates, although disturbed by the rotation, shows a net translation to the left and to counteract this flow in a closed channel, a rightward flow emerges in the front region. The resulting flow profile is shown in Figure 2.

The goal of this project is to emulate the effects shown in the experiment above in simulation.

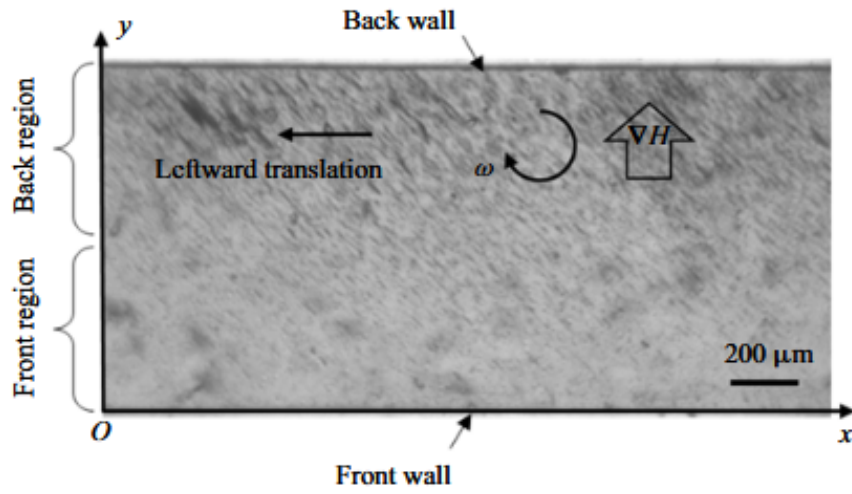


Figure 1: Captured image of system with rod shaped colloidal aggregates in a channel of carrier fluid, shown in Queiros Campos (2024) [1].

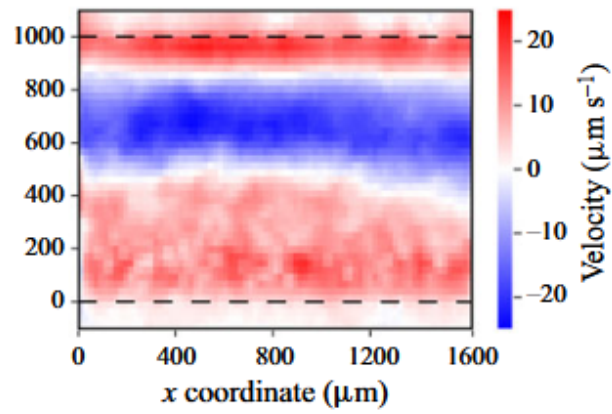


Figure 2: Induced flow profile inside the channel, from Queiros Campos (2024) [1].

2 Model and Methods

2.1 Ellipsoid raspberries

There isn't a physical model underlying the creation of the raspberries. These are merely algorithms to efficiently find configurations.

2.1.1 Smooth shell configurations

The first implemented approach has a dynamic number of particles it tries to maximize, in a specified size of ellipsoid. It also guarantees a surface that is as densely populated as possible. This is motivated by the intention to have these particles emulate a solid within a solvent.

It performs two steps. First, it populates the surface and then the interior defined by the previous step.

A number of particles given by an upper bound estimate for how many will fit are positioned on the surface in a random configuration. They interact via a Weeks-Chandler-Anderson potential. The system is iterated at very low temperature, with the particle positions being rescaled to the ellipsoid surface after each step. This will result in a tightly packed surface with many overlaps between particles. The mean distance between next neighbours is tracked, as well as the smallest distance between any of those pairs. The target for both being to reach 1. Once neither metric improves for a set amount of time, a particle is removed, to allow a looser configuration. This is repeated until a configuration is found where no overlap exists anymore and the particles are then fixed in space. There are several optimizations involved, such as setting a higher temperature for a brief time after particle removal to equalize faster, and only checking thoroughly for suitable configurations after the tracked values are suitably close to the target.

To fill the interior, another number of particles is positioned inside the ellipsoid bound, in a hexagonal packing, interacting with each other and the outer shell and then iterated. This leads to strong forces that push particles through the outer shell. After a certain amount of force is not exceeded for a set time anymore, all particles outside the ellipsoid are removed and the rest continue to try and find a suitable configuration after the same method as the shell.

2.1.2 Homogeneous configurations

2.2 Thermal Stoner-Wohlfarth

2.2.1 Model

The one serious point I can think of for now.

Go to source of [2] for model for thermalized stoner wohlfarth model.

2.2.2 Method

Mostarac 2025 [2]

2.3 Computational methods, not Espresso specific

Basically explain the parts that I do with Espresso, without being specific to Espresso.

3 Results and discussion

Credit use of ESPResSo [5] [6] [7].

3.1 Ellipsoid setup

Quantitative stuff. Mean distances and variances.

Despite a good amount of time spent optimizing, the smooth shell algorithm remains gruelingly inefficient due to the poor conditioning of the problem. The nature of the WCA potential in combination with the number of neighbour pairs leads to massive amounts of computational time being wasted on persistent, tiny overlaps. The filling of the inside is simpler, but produces large gaps, especially on thin ellipsoids.

Other method here

In summary, the homogeneous algorithm performs significantly faster, produces a smooth enough surface to be comparable, and achieves tighter overall packing.

3.2 else(?)

Showing that the parameters and the susceptibility show roughly the expected relationship, as in the presentations is easy, but lame.

??? actually validating results against something sure would be nice. Not sure how to do that yet.

Validating the susceptibility dependent on the semiaxes ratio should be fairly easy.

Calculate theoretical shape anisotropy energy from $E = 1/2\mu_0 Ms^2(Na - Nc)\sin^2(\theta)$ (source to be checked: Osborn, J. A. (1945)) and compare to simulation results of high field over all angles at low temperature. But that only makes sense if I set all easy axes in line with the principal axis of the ellipsoid.

Should recreate SW, have done this before. That was with the interesting caveat that switching field is $1/3$ at 45° instead of $1/2$. That was probably due to the particle easy axes being randomly oriented...

Emergent shape anisotropy of the ellipsoid being dependent on the distribution of the internal ones is funky...

Not sure how to validate anything more complex...

!!! In general: I lack a comprehension of what the system is, depending on the way the easy axes are configured. Random directions: superparamag, net zero mag due to thermals, but high susceptibility, no hysteresis aligned directions: ferromag, overpowers thermal energy at 0 field, has net mag and hysteresis. I guess actually drawing a full hysteresis might be valuable.

I think I might be thinking about that too rigidly either way. Paper states susceptibility of 22 and in the same breath calls the aggregates paramagnetic?

4 Conclusion

4.1 Summary of findings

4.2 Outlook

References

- [1] J. Queiros Campos, M. Raboisson-Michel, S. Schaub, S. Toe, L. Boulant, G. Verger-Dubois, A. Zubarev, and P. Kuzhir, “Field-induced macroscopic flow of a dilute self-assembling magnetic colloid under rotating magnetic fields,” *Journal of Fluid Mechanics*, vol. 981, Feb. 2024.
- [2] D. Mostarac, A. A. Kuznetsov, S. Helbig, C. Abert, P. A. Sánchez, D. Suess, and S. S. Kantorovich, “Thermal stoner-wohlfarth model for magnetodynamics of single domain nanoparticles: Implementation and validation,” *Physical Review B*, vol. 111, p. 014438, Jan. 2025.
- [3] X. Chen, C. Zhou, and W. Wang, “Colloidal motors 101: A beginner’s guide to colloidal motor research,” *Chemistry – An Asian Journal*, vol. 14, pp. 2388–2405, May 2019.
- [4] R. Cheng, W. Huang, L. Huang, B. Yang, L. Mao, K. Jin, Q. ZhuGe, and Y. Zhao, “Acceleration of tissue plasminogen activator-mediated thrombolysis by magnetically powered nanomotors,” *ACS Nano*, vol. 8, no. 8, pp. 7746–7754, 2014.
- [5] R. Weeber, J.-N. Grad, D. Beyer, P. M. Blanco, P. Kreissl, A. Reinauer, I. Tischler, P. Košován, and C. Holm, *ESPResSo, a Versatile Open-Source Software Package for Simulating Soft Matter Systems*, pp. 578–601. Elsevier, 2024.
- [6] F. Weik, R. Weeber, K. Szuttor, K. Breitsprecher, J. de Graaf, M. Kuron, J. Landsgesell, H. Menke, D. Sean, and C. Holm, “ESPResSo 4.0 – an extensible software package for simulating soft matter systems,” *The European Physical Journal Special Topics*, vol. 227, no. 14, pp. 1789–1816, 2019.
- [7] J.-N. Grad, F. Weik, A. Reinauer, H. Kobayashi, I. Tischler, P. M. Blanco, D. Mostarac, S. Bindgen, D. Beyer, J. Hoßbach, M. Kuron, P. Hohenberger, N. Müller, R. Frenner, Y. Gandhi, M. E. Brito, S. Tovey, and R. Weeber, “Espresso,” 2026.